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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

2143443030

PHYSICS **5054/43**

Paper 4 Alternative to Practical **May/June 2010**

1 hour

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **9** printed pages and **3** blank pages.

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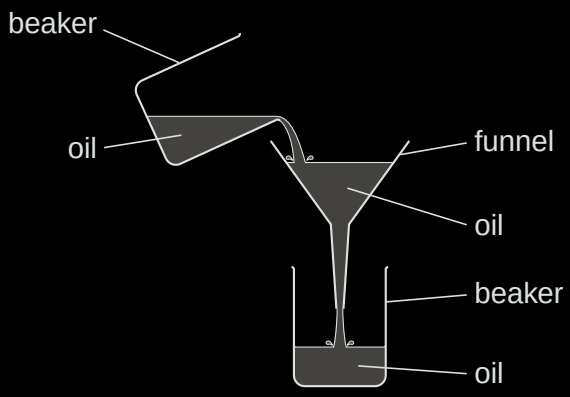


Fig. 1.1

The time *t* taken for the oil to flow through the funnel is recorded with a stopwatch.

(a) State why it is important to stir the oil during heating.

.....

..... [1]

(b) Explain why the oil is heated **gently**.

.....

..... [1]

(c) The experiment is repeated for several values of θ. Values of θ and *t* are recorded in the table of Fig. 1.2.

θ / °C	*t* / s
69	13.69
52	14.97
35	17.34
24	21.16
17	25.16
10	28.85

Fig. 1.2

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Fig. 1.3 [4]

(ii) Estimate the time taken for oil at 80 °C to flow through the funnel.

..... [1]

(iii) State the maximum reading on a standard laboratory liquid-in-glass thermometer that is suitable for this experiment.

..... [1]

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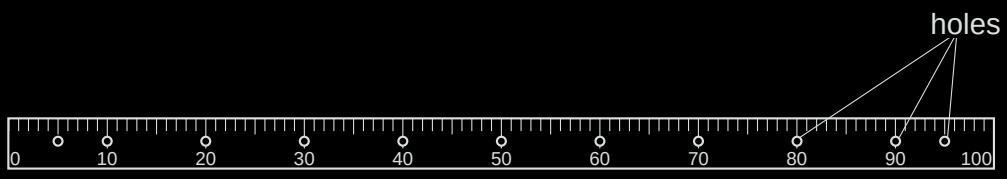


Fig. 2.1

The metre rule is suspended from the hole at 5.0 cm so that it can swing freely, as shown in Fig. 2.2.

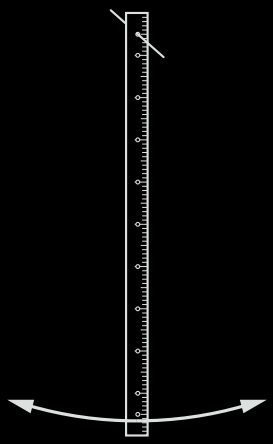


Fig. 2.2

The rule is made to swing from side-to-side and the time T for one complete swing is determined.

(a) Describe three experimental techniques used to obtain an accurate value for T.

1. .....

.....

2. .....

.....

3. .....

.....

[3]

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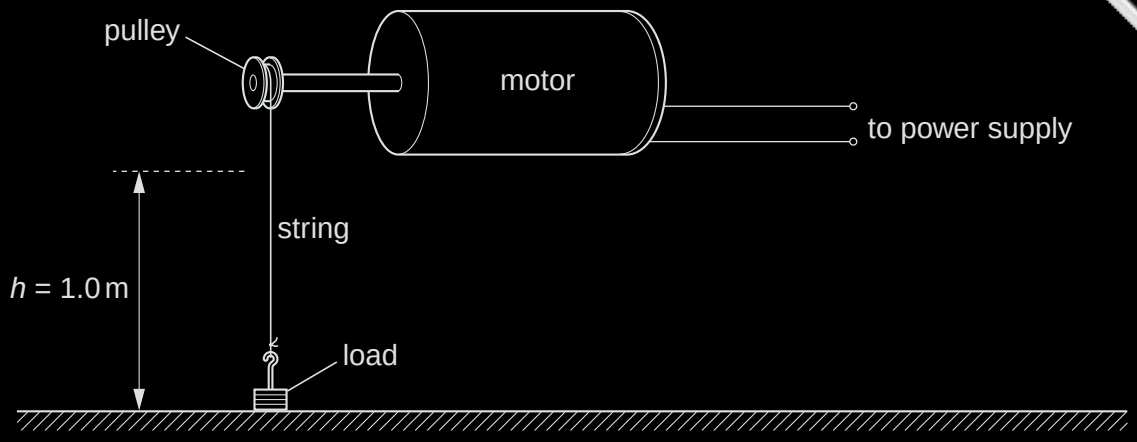


Fig. 3.1 (not to scale)

When the motor is switched on, the string winds round the pulley and the load is lifted.
Six students measure the time t taken for the load to be lifted a height h of 1.0 m.

(a) The times recorded are

4.47 s, 4.53 s, 4.39 s, 4.44 s, 5.92 s, 4.61 s.

(i) The result of 5.92 s is discarded.

Suggest why this result is very much larger than the others.

..... [1]

(ii) Calculate the average time t_{av} of the other results.

Considering the variation in the students' results, give your answer to an appropriate number of significant figures. Show your working.

t_{av} = ... s [2]

(b) The efficiency E of the motor is given by the relationship

$$E = \frac{2 \times 10^2}{3 \times t_{av}} \%.$$

Use this relationship to calculate E.

E = ... [1]

(c) Describe one way in which the students can measure h accurately.

.....

.....

..... [1]

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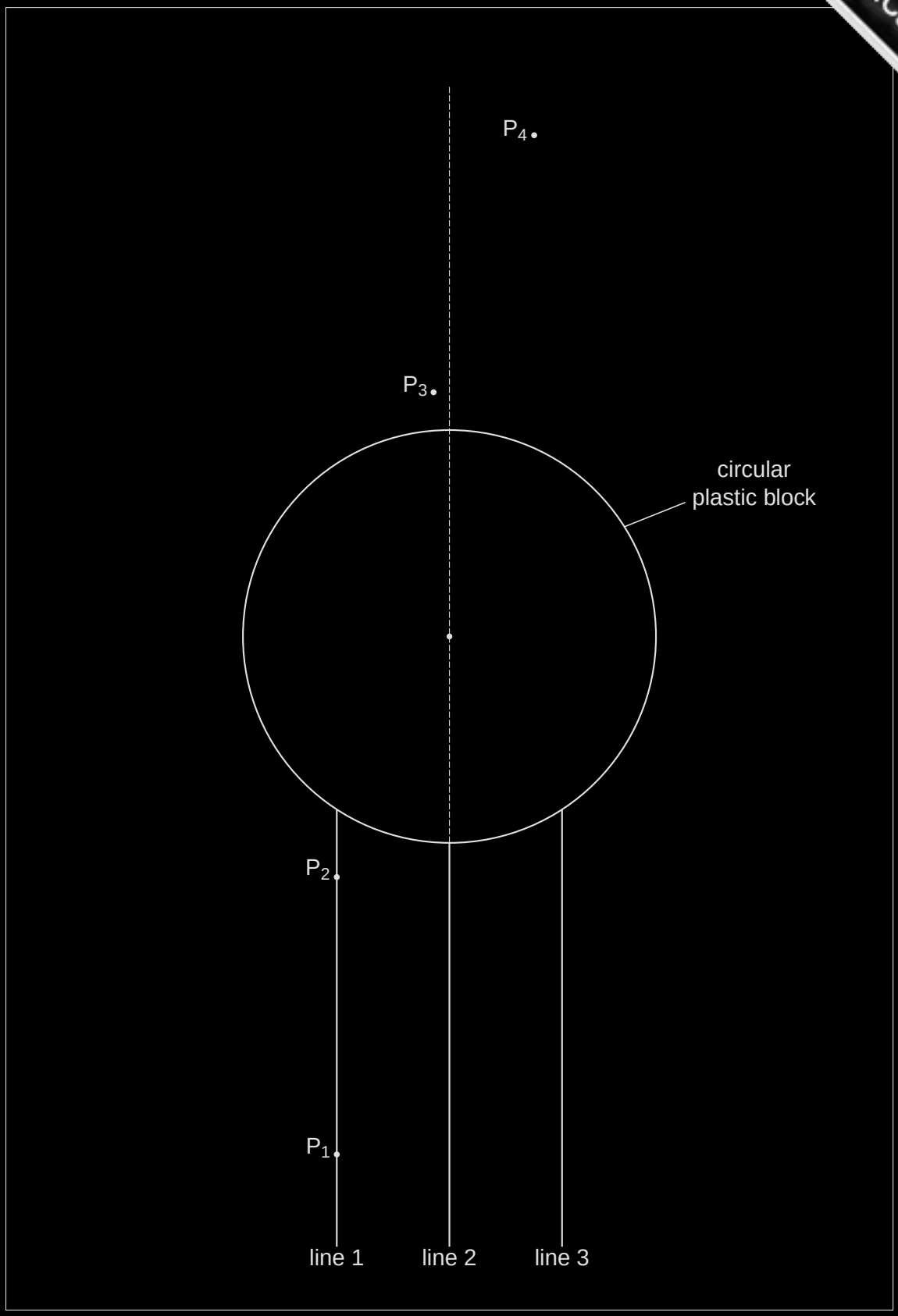


Fig. 4.1

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